Zeckendorf's theorem (8.29)

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Prove **Zeckendorf's theorem** : *Any positive integer N can be expressed uniquely as a sum of distinct Fibonacci numbers containing no neighbors* :

$$N = \sum_{j=1}^{m} F_{i_j+1}, \qquad |i_j - i_{j-1}| \geq 2.$$

Here $F_1 = 1$, $F_2 = 2$, $F_{n+2} = F_{n+1} + F_n$, $n \geq 1$. Indeed, $1 = F_1 = 1$, $2 = F_2 = 10$, $3 = F_3 = 100$, $4 = F_3 + F_1 = 101$, $5 = F_4 = 1000$, $6 = F_4 + F_1 = 1001$, $7 = F_4 + F_2 = 1010$, $8 = F_5 = 10000$, $9 = F_5 + F_1 = 10001$, $10 = F_5 + F_2 = 10010$, $11 = F_5 + F_3 = 10100$, $12 = F_5 + F_3 + F_1 = 10101, \ldots$.